



RICE UNIVERSITY

AEGIS

*A Data-Driven Framework for
NFL Injury Prevention*

Lucca Ferraz, Savan Patel
Team SuperbOwl
Rice University





Table of Contents

01

Introduction

02

Data

03

Methodology

04

Results

05

Discussion

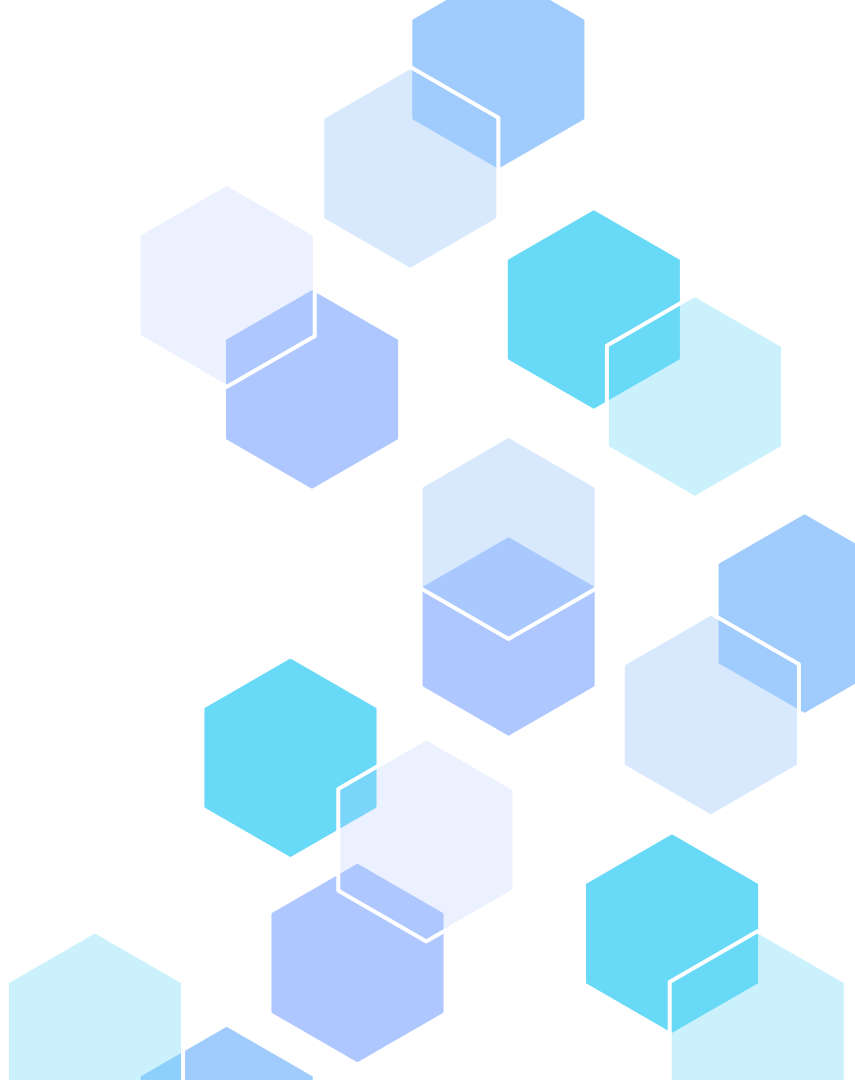
06

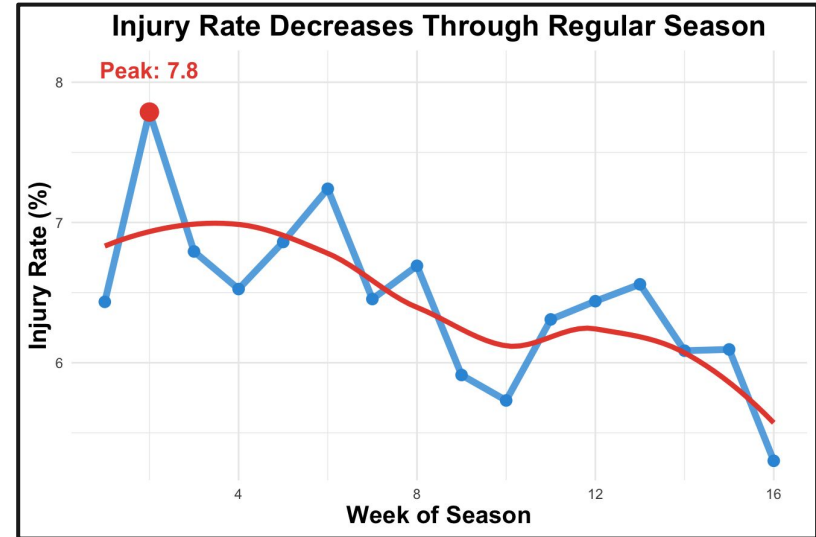
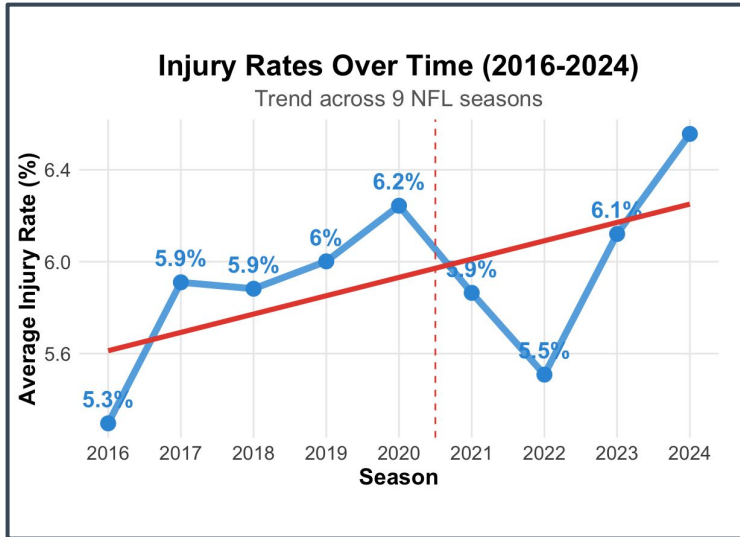
Conclusion



01

Introduction





*Injuries are increasing year after year, but they **decrease** as the season progresses and snap counts increase.*

This presents a unique challenge and opportunity.

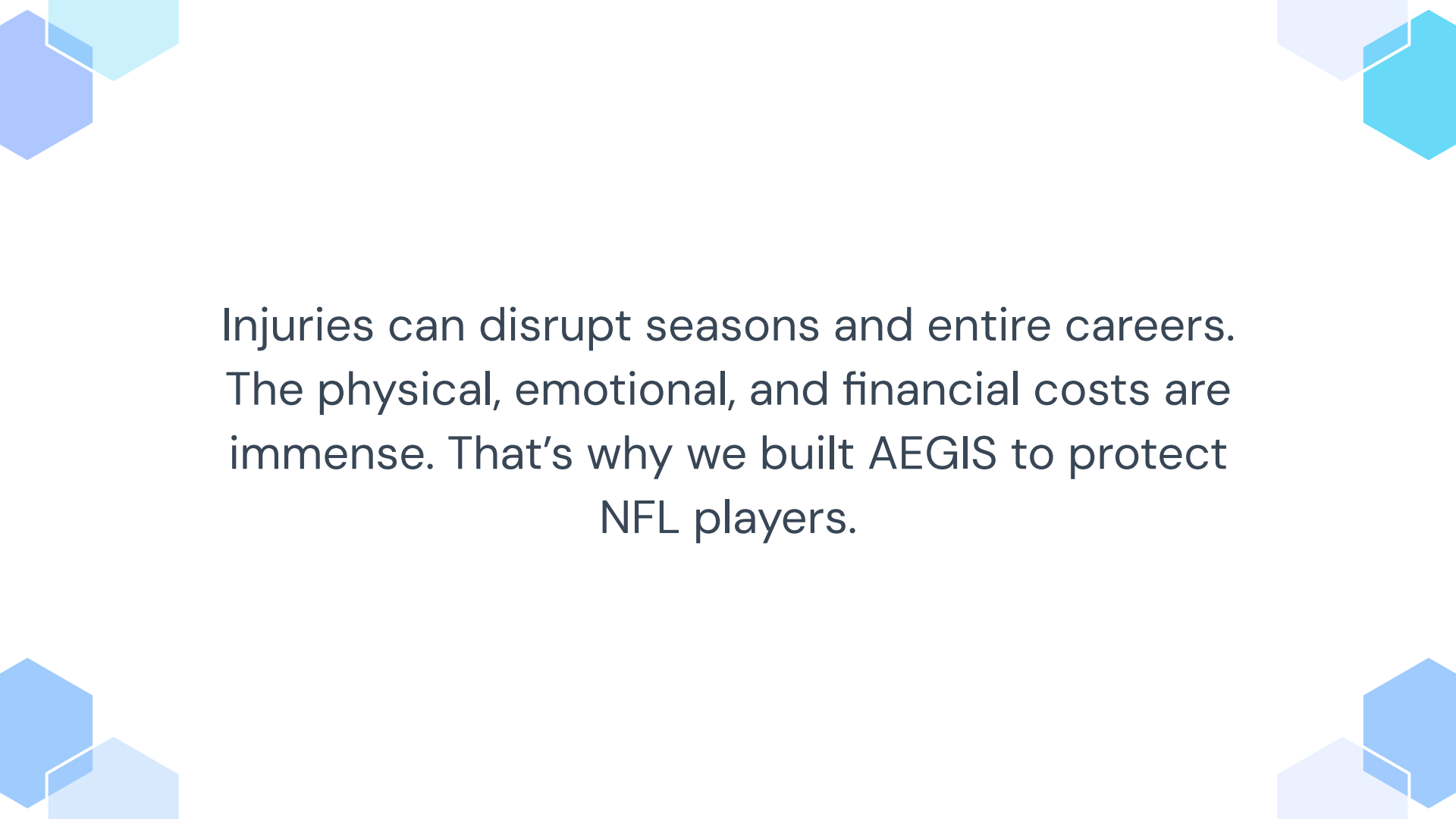
Case study: Deebo Samuel

- In 2024, San Francisco 49er Deebo Samuel was coming off of a 1000-scrimmage yard season
- In week 2 against the Vikings, he had 8 receptions for 110 yards
 - However, he suffered a calf strain, which prevented him from playing the next week against the Rams
- The 49ers would lose 24-27, an important loss against a division rival

It's possible that Samuel's presence could have changed the outcome against the Rams.

Was there any way to predict this soft tissue injury and protect Samuel from unnecessary wear-and-tear?



The slide features a light blue background with decorative hexagonal shapes in the corners. In the top-left corner, there are two overlapping hexagons, one light blue and one medium blue. In the top-right corner, there are two overlapping hexagons, one light blue and one medium blue. In the bottom-left corner, there are two overlapping hexagons, one medium blue and one light blue. In the bottom-right corner, there are two overlapping hexagons, one medium blue and one light blue.

Injuries can disrupt seasons and entire careers.
The physical, emotional, and financial costs are
immense. That's why we built AEGIS to protect
NFL players.

AEGIS

- Aegis: a divine shield associated with Zeus and Athena
- Represents protection, backing, or support of a particular person or organization
- Our metric AEGIS (***A**cute **E**ndurance **G**uidance for **I**njury **S**afeguarding*) aims to protect NFL players from the physical, emotional, and financial harm from soft-tissue injuries
- A nod to the NFL's logo, also a shield, embodying strength, stability, and protection

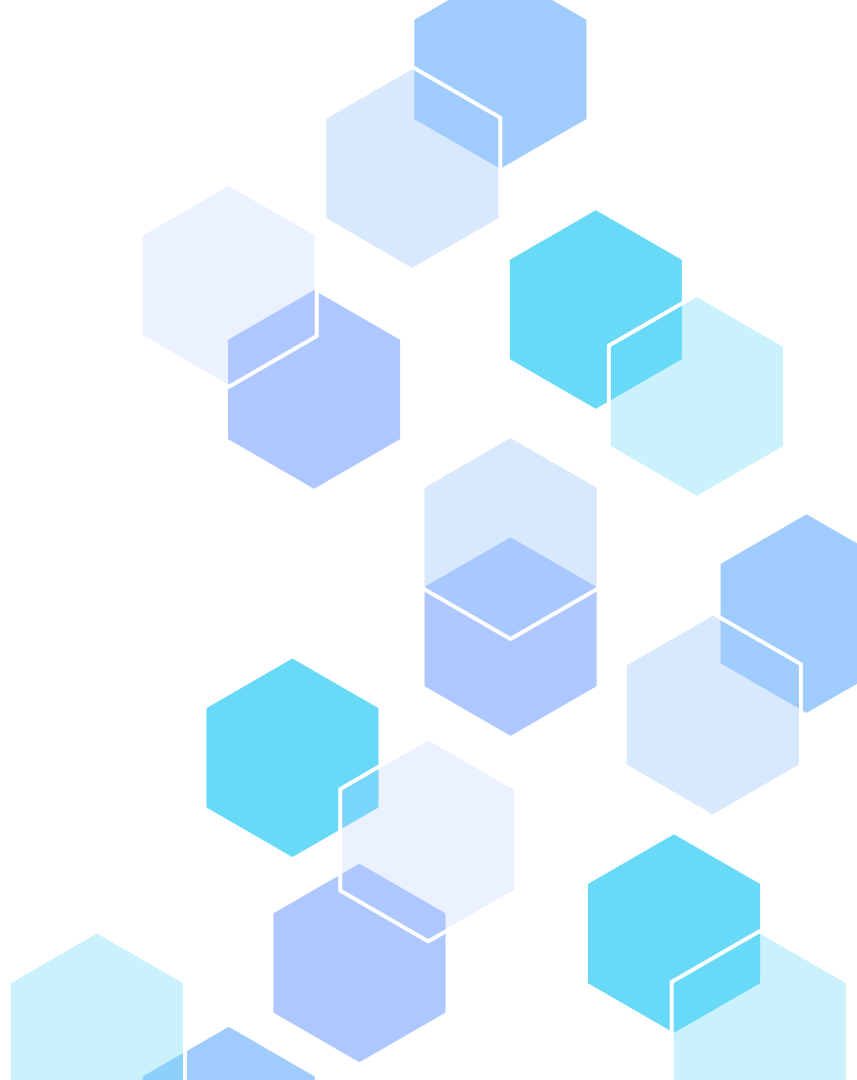


What is AEGIS?

- Biomechanics-inspired predictive model
 - Built to identify when players are at high risk of soft-tissue injuries
- Soft tissue injuries account for 70% of all football-related visits to physical therapists
- Examines three biomechanical stress categories: deceleration events, collisions, and explosive plays, to figure out what drives these injuries at the NFL level

01

Data



Data

Sourcing

- NFLreadR package in R

Data

- Play-by-play data from the 2016–2024 seasons
- 435,483 plays, including outcomes, involved players, and situational context
- Weekly NFL injury reports
- Next Gen Stats' tracking data was ideal, but unavailable for the study

Data Cleaning

- Filtered injury sports to focus exclusively on soft-tissue injuries
 - Exclusively included diagnoses of hamstring, calf, groin, quadriceps, and non-contact knee, achilles, and ankle injuries
- Included only players that have participated in 20% or more of the team's' offensive or defensive snaps
- Player identification handling
 - Merged datasets of rosters, plays, and injuries using unique player IDs and team-week keys
- Created the target variable of “injury in next week” as a binary

Data Processing

- **Biomechanical Proxies (Play-Level)**
 - Deceleration Events
 - Explosive Plays
 - Collision Events
 - Players are only credited if they are involved in the play
- **Metrics:**
 - Acute Deceleration Load
 - Chronic Deceleration Load
 - Acute-Chronic Workload Ratio (ACWR): $\text{acute} \div \text{chronic load}$
 - Stress Accumulation Score

Decelerations

- Eccentric contractions (Sudden stops) are 1.5–3.0x more forceful to soft tissue than concentric contractions
- Identified specific plays as deceleration events
 - Tackles for loss
 - Deep incompletions (>15 air yards)
 - Deep completions, low YAC (>15 air yards and <2 YAC)
 - Safeties



Explosive Plays

- Explosive movements leave athletes at higher risk of soft tissue injury
- Identified specific high-velocity plays
 - Runs >12 yards
 - Receptions >16 yards
 - Plays with Expected Points Added (EPA) >1.5



Collisions

- High-impact collisions generate peak force & direct trauma
- Identified specific collisions during plays
 - Sacks & QB hits
 - Tackles (solo & assisted)
 - Forced fumbles
 - Receivers/RBs upon tackle

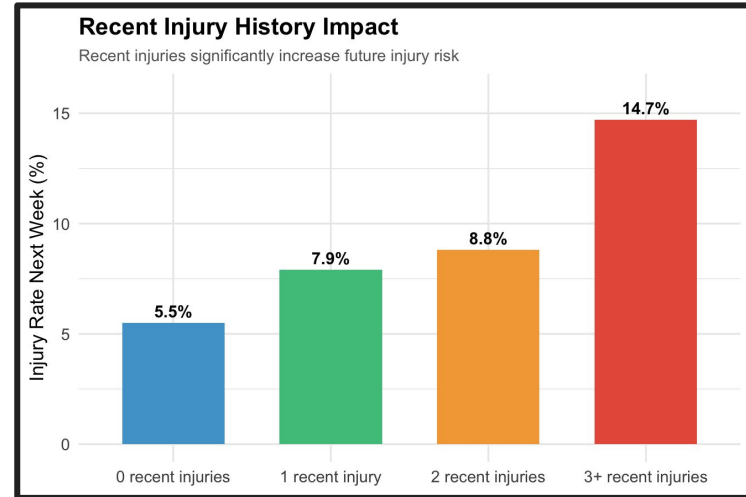


Acute-Chronic Workload Ratio (ACWR)

- Acute Workload: workload in the past seven days
- Chronic Workload: average weekly workload over past 28 days
- ACWR: ratio between acute and chronic workload
- We calculate ACWR for decelerations, explosive plays, and collisions
- Higher ACWR indicates a workload spike, while an $ACWR < 1$ indicates a decline

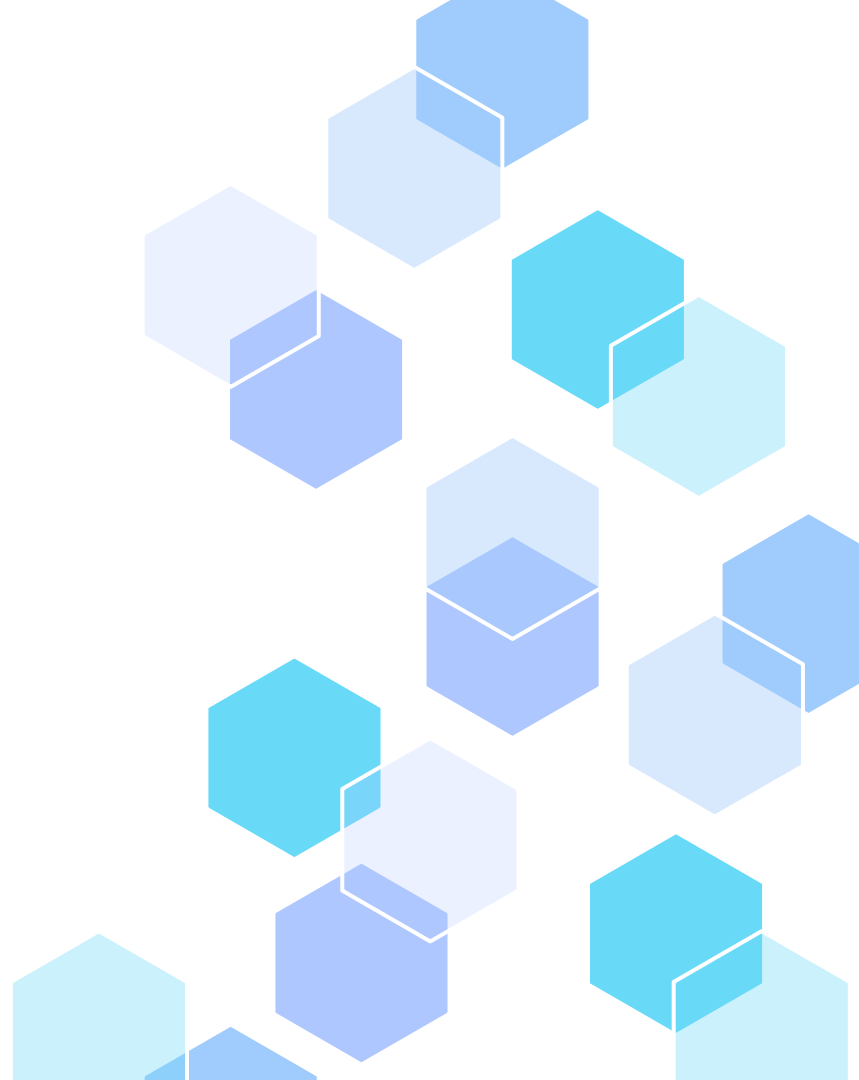
Stress Accumulation

- After an injury, players are more at-risk for another
- Workload while recovering is especially important
- Stress Accumulation: number of injuries over past 4 weeks * ACWR
- Recent vulnerability and current exposure lead to higher risks



03

Methodology



Methodology Workflow



**Data, Feature
Engineering**



**Predictive
Modeling**



**Causal
Analysis**

Statistical Methodology

- **Predictive Modeling (XGBoost Classifier)**
 - Forecasts probability of soft-tissue injury in the following week based on current-week biomechanical, contextual, and historical features
 - XGBoost handles non-linear relationships and interactions between variables
 - Provides clear feature importance to identify key risk drivers.
 - Alternatives considered (and rejected):
 - logistic regression (too simplistic),
 - simpler tree models (lower performance)
 - neural networks (less interpretable)

Statistical Methodology

- **Class Imbalances**

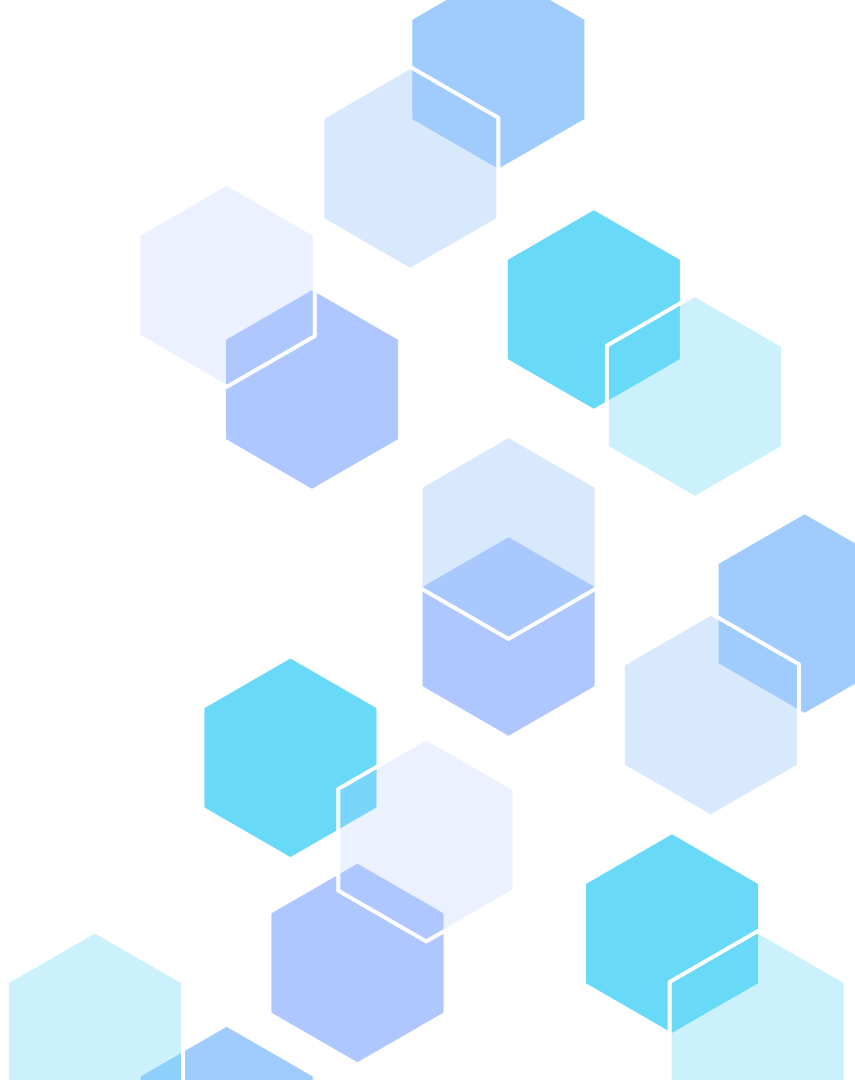
- Our dataset of injuries is extremely unbalanced (< 6% of observations)
- A model that never predicts injuries would perform well using traditional evaluation metrics but always fail to catch actual injuries, missing the objective
- Adjusted using `scale_pos_weight` parameter in XGBoost
- Encourages model to predict injuries
- Makes for better out of sample injury predictions

Statistical Methodology

- **Causal Inference (Propensity Score Matching)**
 - Isolates the independent effect of high deceleration exposure from confounding variables (position, snap volume, injury history).
 - Mimics a randomized experiment by matching high-exposure players with similar low-exposure counterparts.
 - Directly addresses key confounders intrinsic to football.
 - Strengthens scientific evidence for policy recommendations.
 - Alternatives considered (and rejected):
 - Regression adjustment alone (requires perfect model specification)
 - Instrumental variables (we can't randomize treatments)

04

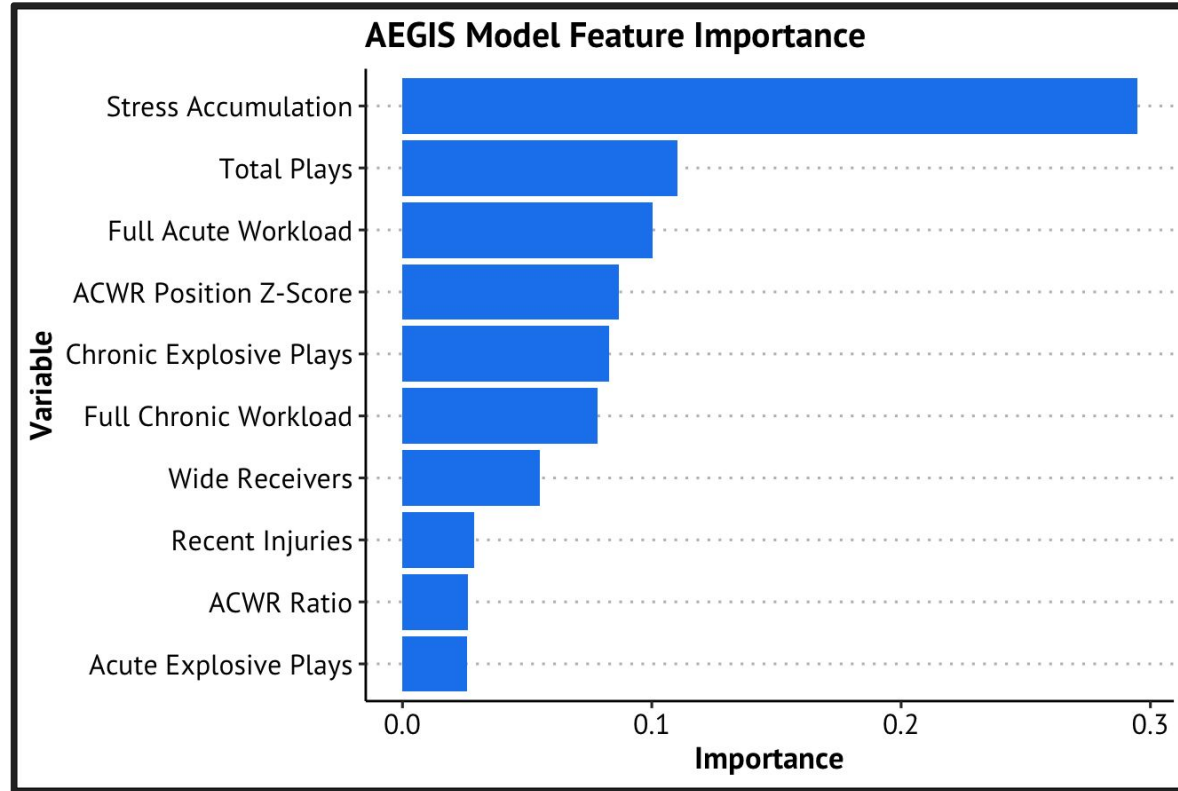
Results



AEGIS Model Results

- Evaluated model using Area Under Curve – Precision Recall
- More informative for highly imbalanced datasets
- All performance metrics calculated out of sample
- Compared model results to assigning the baseline 5.9% injury chance to all predictions
- Our model showed 44% improvement over baseline in AUC-PR
- Correctly identified 57.6% of all injuries in next week

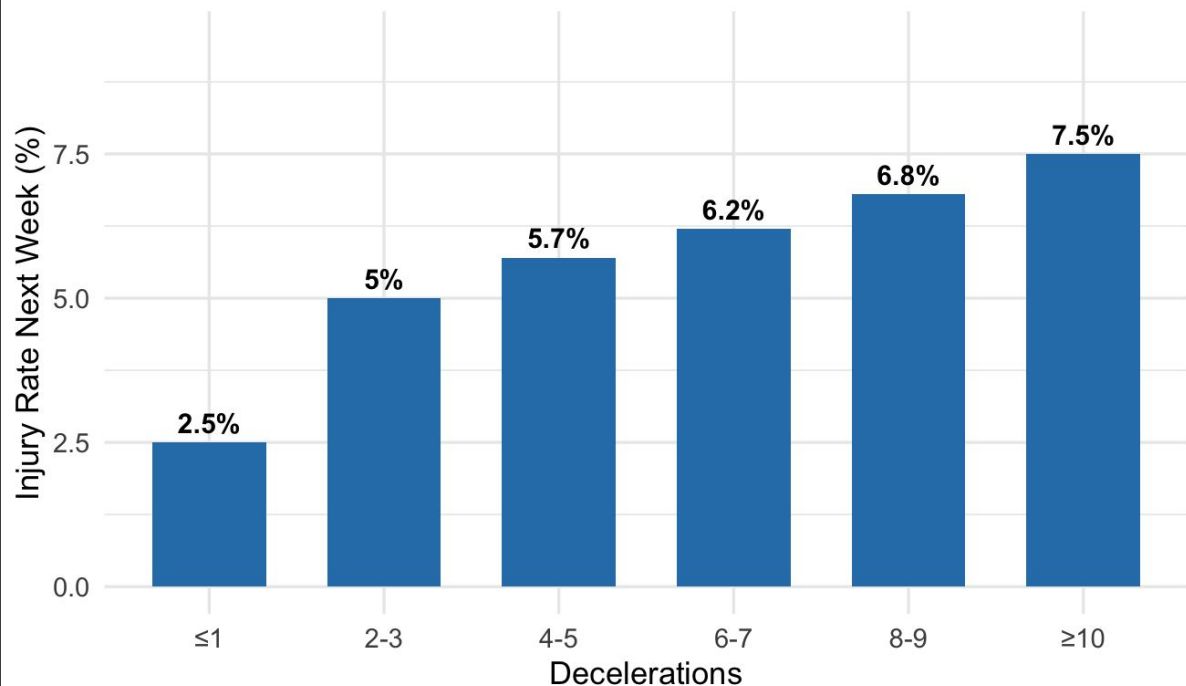
Feature Importance



Causal Inference

Clear Trend: More Decelerations = Higher Injury Risk

Steady increase in injury probability with each deceleration event

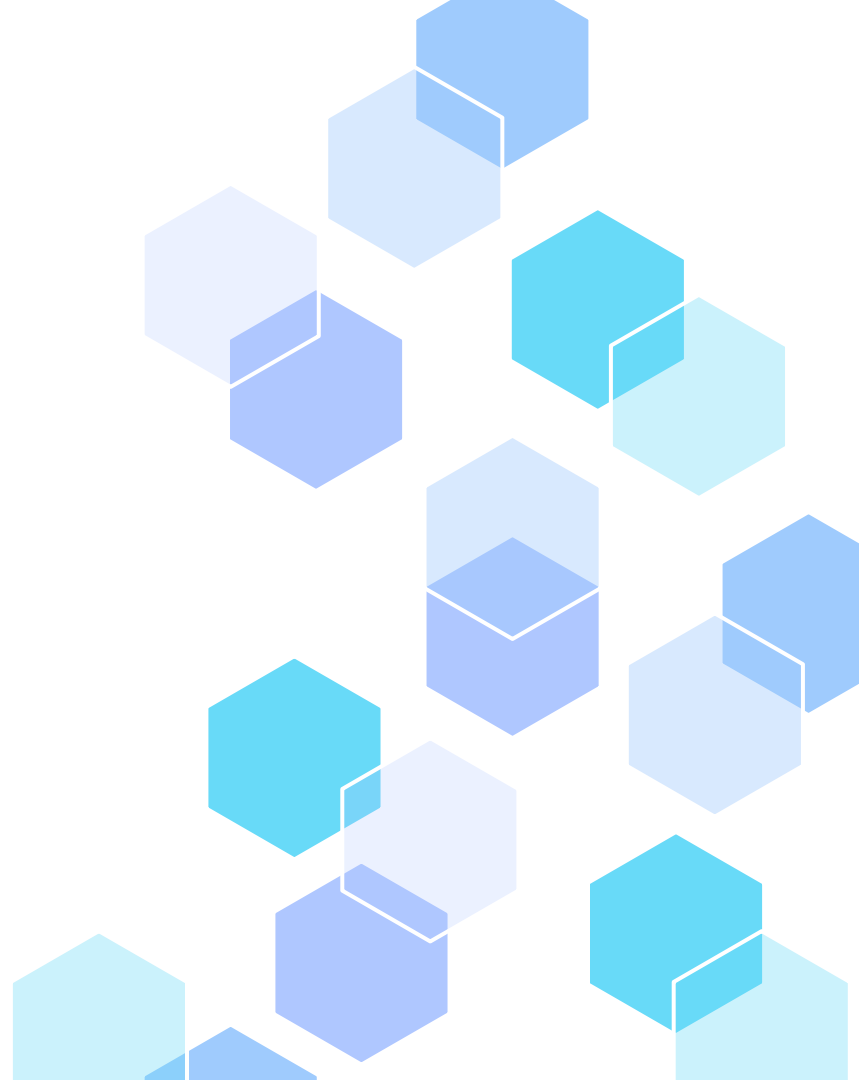


Causal Inference

- Matched players on position, snap volume, and injury history
- High-deceleration exposure was associated with a 0.72% absolute increase in injury probability
 - 95% confidence interval spanned from -0.27% to +1.71%
- This corresponds to a **12% higher relative risk**
- For every 139 player-weeks, high deceleration exposure would be expected to produce one additional injury

05

Discussion



CBA Recommendations

CBA-mandated position-specific safety standards

If players are exceeding safety standards, they can have the option to have reduced workloads in practices.

Establishment of Biomechanical Data Resumes for free agency

NFLPA can create reports of players' historical loads, resilience, and any past safety threshold violations by former teams, ultimately for use by players and agents in negotiations.

CBA Recommendations

Utilization of AEGIS or a similar framework for Biomechanical Health Guarantees in the CBA

If players are injured in weeks of significant excession of safety thresholds, salary protection can be an option.

League-wide load management alerts

Automated alerts to players and the NFLPA, allowing players to advocate for their health to their team's staff.

Impacts

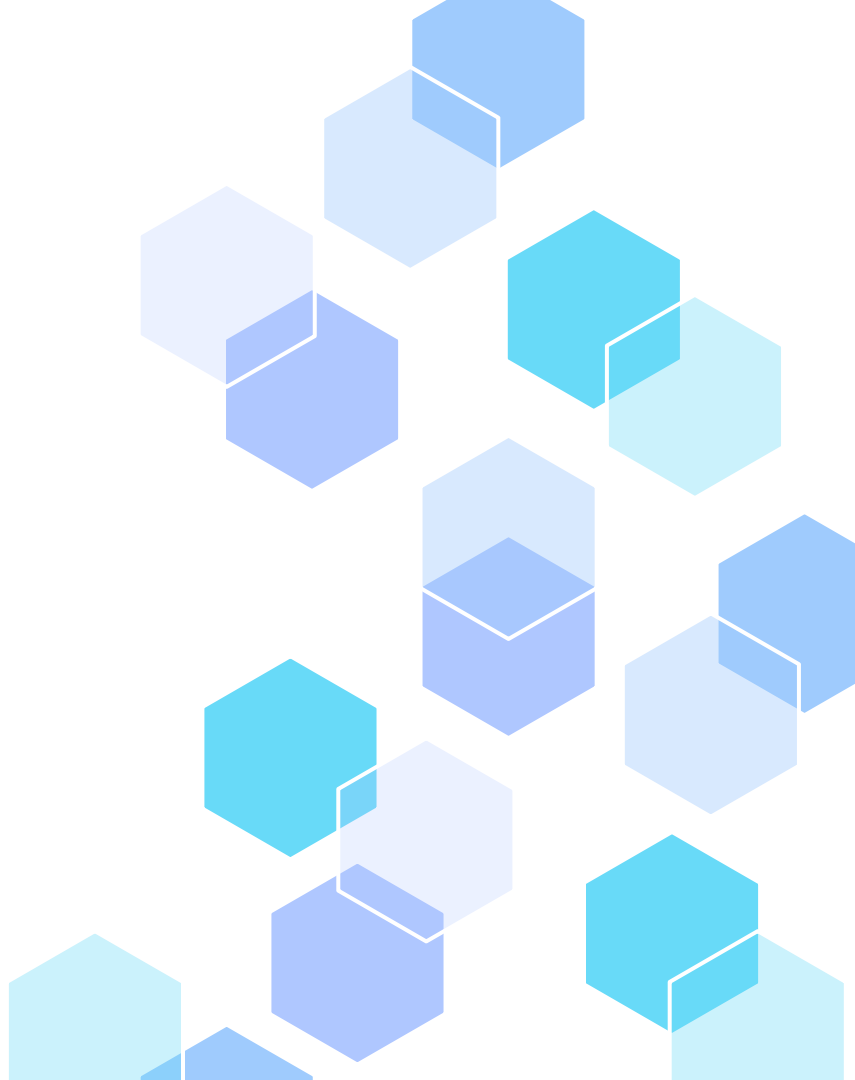
- **Projected 10–15% reduction in soft-tissue injuries**
 - Based on a 12% higher relative risk of injury from deceleration exposure
- **Potential career extension of 1–2 seasons**
 - If a player can avoid a catastrophic career-altering injury and several minor ones, an extension of their high-performance career window is reasonable
 - Dependent on the player: some players are anatomically more predisposed to tissue injuries
- **Contract Negotiation Tools**
 - Agents and players can show teams that a previous injury may have been based on overuse rather than the player's qualities
 - Negotiation of premium pay or guarantees

Next Steps / Future Research

- Integrate tracking data for true biomechanical movement
 - True measures of deceleration, velocity, collision force
- Research on physiological injury risk predictors
 - I.e, limb length, height/weight, bone density, strength, speed
- If implemented, AEGIS or a similar framework could become an athlete-facing digital dashboard allowing players and the NFLPA to access their load and injury data

05

Conclusion



Wrap-up

- Soft-tissue injuries are frequent (and increasing) in the NFL and can alter careers and game outcomes, yet many are preventable with better workload insight
- We introduce AEGIS, biomechanics-inspired predictive model that identifies high injury risk using deceleration, explosive play, and collision data
- AEGIS improves injury prediction by 44% and shows a 12% higher injury risk from high deceleration exposure
- If adopted, we believe AEGIS could reduce soft-tissue injuries by 10–15%, extend careers, and empower players with data for health and contract decisions
- Integration of tracking data and league-wide safety alerts could further advance injury prevention in the NFL

Thank you!

Any questions?

